

Riccardo Corte

MSc Student — Physics of Data

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Profile

I would describe myself in one word: *grit*. I am a Physics graduate and MSc student focused on machine learning and data-driven methods, with particular interest in transformer architectures, differential geometry, and representation learning.

I approach problems by combining strong logic intuition with practical implementation and rigor, building clear strategies and following them through complex technical challenges. My work is driven by a strong interest in understanding how models behave, not only in applying them.

Education

MSc Physics of Data

2024 – Present

University of Padova, Italy

Focus on machine learning, large-scale data analysis, and transformer architectures.

BSc Physics — 97/110

2021 – 2024

University of Trieste, Italy

Thesis on optical aberrations using ray tracing and analytical modeling. Erasmus at Universidad de Granada.

Student association president; elected to department council; interviewed Nobel Prize winner Giorgio Parisi.

Selected Projects

Multilingual Fine-Tuning of Language Models

Flagship Project

- Built a controlled fine-tuning and evaluation pipeline for a 3B instruction-tuned transformer (Qwen2.5-3B-Instruct), targeting multilingual question answering.
- Implemented parameter-efficient fine-tuning on curated multilingual datasets (IT/FR/EN), with probe languages (ES/PT) to evaluate cross-lingual generalization.
- Trained with LoRA under single-GPU constraints (NVIDIA T4), optimizing for memory efficiency and reproducibility.
- Designed a baseline vs fine-tuned comparison framework (exact match, semantic similarity, response length, failure analysis).
- **Observed a consistent semantic improvement on the main test set** (241→289 successes under embedding similarity; posterior $p(\text{FT} > \text{Base}) = 0.997$).
- **Identified a controlled “attribute shift”**: outputs became shorter and more target-aligned, with moderate transfer to Spanish and weak evidence of improvement on Portuguese.

LLaMA Inference from Scratch & Bayesian Analysis

Work in Progress

- Implemented a LLaMA-style transformer from scratch in PyTorch, including grouped-query attention, rotary positional embeddings (RoPE), and KV caching for efficient autoregressive decoding.
- Loaded and executed pretrained LLaMA-2 (7B) weights, enabling token-by-token generation on GPU (Tesla T4) with a custom inference pipeline.
- Designed experiments comparing transformer predictions to analytical Bayesian posteriors on synthetic datasets.
- Investigating implicit priors and hidden-state information content in relation to Bayesian updating.

GPT-2: Transformer Embedding Dynamics

- Analyzed the evolution of embedding representations across transformer layers, focusing on attention and feed-forward (FFNN) transformations.
- Studied hidden-state geometry using dimensionality reduction (PCA, t-SNE) to track structural changes across depth.
- Identified systematic patterns in representation dynamics and limitations in interpreting high-dimensional model behavior.

GPU-Parallel Distributed QR with Dask

- Implemented distributed QR decomposition algorithms (Cholesky, indirect, TSQR) using Dask on a compute cluster.
- Benchmarked scalability across matrix size, partitioning, number of workers, and conditioning on large-scale datasets.
- Explored trade-offs between numerical stability and parallel efficiency, including GPU-accelerated execution.

Mean-Reversion Trading Bot — XSOR Capital

- Developed a quantitative metric (η) to characterize mean-reverting vs trending behavior in financial time series.
- Implemented complementary estimation methods (autoregressive modeling and statistical sampling) with rolling-window extension.
- Built an end-to-end pipeline for signal generation and strategy evaluation.

Research Experience

Research Internship — Precision Engineering & Time Synchronization

2023

- Contributed to a technical-historical study on precision clock synchronization systems.
- Co-authored a published book chapter on 20th-century engineering methods and timekeeping accuracy.

Skills

Machine Learning & AI

Transformers, LLM fine-tuning (LoRA), representation learning, neural networks (CNN, RNN), RBMs

Programming

Python (advanced), PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, C++, R

Systems & Infrastructure

Linux, Bash, Docker, distributed computing (Dask), GPU workflows, SSH, Git

Modeling & Mathematics

Bayesian inference, differential geometry, statistical modeling, time series analysis, optimization, stochastic processes, linear algebra

Languages

English (C1 — Cambridge CAE), Spanish (B2), German (basic), French (basic)

Awards

Italian Logic Games — Silver Medalist (2018)

Team captain and individual finalist for four consecutive years (best placement: 4th)

Interests

Endurance sports (Rome Marathon 2025, multiple half-marathons), chess, tennis

Links

Complete project reports, experiments, and additional results are available at:

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